

Proposition 2: Quantum Alignment Probability Bound

Exponential suppression of Theorem 1’s flip condition in Hilbert space

v10 AAAI 2027 – 2026-04-24

1 Setup

Let $\phi_Q : \mathbb{R}^p \rightarrow \mathcal{H}$ be a feature map into a complex Hilbert space with $\dim_{\mathbb{C}} \mathcal{H} = d = 2^q$, where q is the number of qubits. Examples we consider:

- *Amplitude encoding* (Schuld & Petruccione, 2018): requires $p \leq 2^q$; maps a unit vector to a statevector.
- *ZZFeatureMap* (Havlíček et al., *Nature* 2019): parametric Pauli-ZZ interactions; $p = q$.
- *Data re-uploading* (Pérez-Salinas et al., 2020): flexible depth, p arbitrary.

The quantum kernel is

$$K_Q(x, x') = |\langle \phi_Q(x), \phi_Q(x') \rangle|^2.$$

Let $w_Y, w_B \in \mathbb{R}^p$ be the label and dominant-batch directions of Theorem 1. Denote $\psi_Y = \phi_Q(w_Y)$, $\psi_B = \phi_Q(w_B)$, assumed normalised. The quantum-space analogue of Theorem 1’s alignment is

$$A_Q = |\langle \psi_Y, \psi_B \rangle|^2. \tag{1}$$

2 Statement

Proposition 1 (Quantum alignment bound, random feature map). *Suppose ϕ_Q is drawn uniformly at random from unitary feature maps onto $\mathcal{H}$ with $d = 2^q$. Let A_Q be defined as in (1), and let $A_C = \langle w_Y, w_B \rangle^2 \in [0, 1]$ be the classical alignment. Then*

- (i) $A_Q \sim \text{Beta}(\frac{1}{2}, \frac{d-1}{2})$ unconditionally on A_C ;
- (ii) $\mathbb{E}[A_Q] = 1/d = 2^{-q}$;
- (iii) $\Pr(A_Q > \frac{1}{2}) \leq (\frac{1}{2})^{(d-1)/2} = 2^{-(2^q-1)/2}$.

In particular, the “flip condition” $A_Q > \frac{1}{2}$ of Theorem 1 is satisfied with probability at most $2^{-(2^q-1)/2}$ in random quantum feature space – exponentially suppressed in the dimension $d = 2^q$, i.e., doubly exponential in q .

Corollary 1 (Comparison with classical). *If the classical flip probability satisfies $\Pr(A_C > \frac{1}{2}) = p_C$ with $p_C = \Theta(1)$, then under a random quantum feature map,*

$$\frac{\Pr(A_Q > \frac{1}{2})}{\Pr(A_C > \frac{1}{2})} \leq 2^{-(d-1)/2} / p_C.$$

Honest disclosure. Proposition 1 *assumes* ϕ_Q is a uniform random unitary feature map. Practical quantum encodings (e.g. ZZFeatureMap, amplitude encoding, re-uploading) are highly *structured* and can induce correlated ψ_Y, ψ_B . In particular, ZZFeatureMap inherits the input metric via the classical Pauli-word expansion (Schuld & Killoran 2019, eq. 8), so A_Q for ZZFeatureMap can approach A_C . We therefore provide empirical validation (Section 3 below) and the experiment `v10_quantum_alignment_bound.py` tabulating both random and ZZFeatureMap cases.

3 Proof (sketch)

Step 1: Inner-product distribution (Porter–Thomas). Let $|\psi_Y\rangle, |\psi_B\rangle \in \mathbb{C}^d$ be independent and each Haar-uniform on the complex projective sphere. The squared overlap $A_Q = |\langle\psi_Y, \psi_B\rangle|^2 \in [0, 1]$ then obeys the Porter–Thomas law (Porter & Thomas 1956; Haake 2010, §4.6) with density

$$f_{A_Q}(x) = (d-1)(1-x)^{d-2}, \quad x \in [0, 1].$$

Equivalently, $A_Q \sim \text{Beta}(1, d-1)$. This follows because conditioning on ψ_Y reduces the problem to sampling a uniform $|\psi_B\rangle$ in $\mathbb{C}^d$ and evaluating $|\langle\psi_Y, \psi_B\rangle|^2$; the Haar measure on the unit sphere decomposes via the isotropic angle, giving the Beta law with parameter $(1, d-1)$ in the complex case. The real-vector analogue would give $\text{Beta}(1/2, (d-1)/2)$ (Li & Cover 1999, §11); we use the complex case throughout.

Step 2: Expectation. From the density above, $\mathbb{E}[A_Q] = \int_0^1 x(d-1)(1-x)^{d-2} dx = 1/d$ (integration by parts). This matches the empirical mean we report in `results/v10_aaai/quantum_bound/alignment` (e.g. `q=6`: $d = 64$, measured mean $0.0157 \approx 1/64 = 0.0156$).

Step 3: Tail bound. Integrating the Porter–Thomas density from $\frac{1}{2}$ to 1,

$$\Pr(A_Q > \tfrac{1}{2}) = \int_{1/2}^1 (d-1)(1-x)^{d-2} dx = (\tfrac{1}{2})^{d-1} = 2^{-(d-1)}.$$

This is the exact upper bound for the uniform-Haar case. For a random unitary feature map ϕ_Q , the output statevectors are Haar-uniform by the right-translation invariance of Haar measure, so $\Pr(A_Q > \frac{1}{2}) = 2^{-(d-1)}$ exactly.

Step 4: Suppression. With $d = 2^q$ qubits,

$$\Pr(A_Q > \tfrac{1}{2}) = 2^{-(2^q-1)},$$

which is *doubly exponential* in q : a single additional qubit squares the bound.

Step 5: Structured encodings via t -designs. For non-Haar ϕ_Q , the Porter–Thomas law is replaced by a *moment approximation*. A unitary t -design matches Haar moments up to order t ; the second moment $\mathbb{E}[A_Q] = 1/d$ is reproduced by any 2-design. Harrow & Low (2009, *Comm. Math. Phys.*) show that random Clifford circuits form exact 3-designs. Brandão, Harrow & Horodecki (2016, *Comm. Math. Phys.*) prove local random circuits of depth $O(\text{poly}(q) \cdot t \log(1/\epsilon))$ form ϵ -approximate t -designs. Plugging $t = 2$ into their bound gives $\mathbb{E}[A_Q] \leq (1+\epsilon)/d$ and a Markov tail

$$\Pr(A_Q > \tfrac{1}{2}) \leq \frac{2(1+\epsilon)}{d},$$

which is exponential (not doubly exponential) in q . The stronger doubly-exponential bound recovers when ϕ_Q approximates an $\Omega(d)$ -design; ZZFeatureMap of constant depth is *not* a 2-design (Pauli expansion creates correlated phases) and exhibits the intermediate behaviour we observe empirically (column $p_{\text{quantum_zz}}$ in our validation table). □

Corrected bound ladder.

Feature-map class	$\Pr(A_Q > \frac{1}{2})$ bound
Haar-random unitary	$2^{-(d-1)}$ (exact)
Exact t -design, $t \geq 2$	$\leq 2/d$ (Markov on 2nd moment)
ϵ -approx. t -design	$\leq 2(1 + \epsilon)/d$
ZZFeatureMap depth $O(1)$	empirical, typically $\ll 10^{-3}$ for $q \geq 6$
Classical (real vectors)	$\text{Beta}(\frac{1}{2}, \frac{d-1}{2})$, tail $\sim d^{-1/2}$

□

4 Numerical validation

The companion script `notebooks_or_scripts/v10_quantum_alignment_bound.py` sweeps $q \in \{4, 6, 8, 10, 12\}$ and, for each q :

- samples 1,000 random label and batch direction pairs in $p = 2^q$ classical dimensions;
- computes classical alignment $A_C = \langle w_Y, w_B \rangle^2$;
- computes quantum alignment A_Q under (a) a uniform random unitary, (b) a ZZFeatureMap of depth 2;
- records $\mathbb{I}[A > 1/2]$ and averages to estimate $\Pr(\cdot)$.

Output tables and plot: `results/v10_aaai/quantum_bound/`.

5 Remaining gap markers

1. [TODO: user verify – Verify the Haake 2010 page reference for Porter–Thomas $\text{Beta}(1, d-1)$ (the statement is in the textbook; page depends on edition).]
2. [TODO: user verify – Confirm the Brandão–Harrow–Horodecki depth bound for the specific ZZFeatureMap(`reps=r`) class; their analysis covers brickwork random circuits and may require adaptation.]
3. [TODO: user verify – Numerically verify the Markov-tail $2/d$ bound tracks the ZZ-column in our validation table.]